EEE LAB-2113: Project Report

Department of Computer Science and Engineering

University of Dhaka

Title: 3-Bit Flash Analog-to-Digital Converter

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1 Introduction

Analog-to-digital converters (ADCs) are the main components of current electronics. They assist in the translation of real-world signals into digital data that computers can understand. ADCs are present in many things like medical devices, car sensors, phones, and more.

Flash ADCs are incredibly fast since they can at once look at all the signal levels. This is done by the comparison of the input signal with already defined levels. Other ADCs that take multiple steps to convert, Flash ADCs do it all in one go. They are highly fast, which is quite useful for fast-moving signals.

The ADC that we are referring to here utilizes the standard components that are proven to be good and easy to locate. The resistor ladder containing 1 k ohm resistors assists in the setting of the levels for comparison. The operational amplifiers (op-amps) are the comparators, they check if the input signal is higher or lower than the reference level from the resistor ladder.

Subsequently, the priority encoder joins the outputs from the comparators to make a 3-bit digital code. Thus, the complexity of the output is reduced and it is moreover more precise by selecting the most similar to the input signal.

This report will give a thorough description of the process of designing and testing a 3-bit Flash ADC circuit. This specific ADC employs such electronic parts as a resistor ladder network, LM324 operational amplifiers, and a CD74HC147 binary encoder to convert analog signals into digital ones with just 3 bits. We test how fast it is, how accurate it is, and how much power it needs to show how good it is for the fast dealing with signals.

2 Theory

In a 3-bit Flash ADC, there are 8 resistors arranged in what's called a resistive voltage divider. These resistors help set up reference voltages that the comparators use for comparison. Each step in this ladder of voltages is figured out using a formula:

$$V_i = \frac{R \cdot V_{ref}}{R + (7-i) \cdot R} \text{ , for } i = 0 \text{ to } 7$$

This gives us a set of reference voltages like this:

$$V_i = \frac{i}{8} \cdot V_{ref} \text{ , where } i \text{ corresponds to the } i\text{th comparator}$$

Now, every comparator in the ADC takes the input voltage V_i and compares it to these

reference voltages. The output of each comparator C_i is either 1 or 0, depending on whether the input voltage is higher than the reference voltage or not.

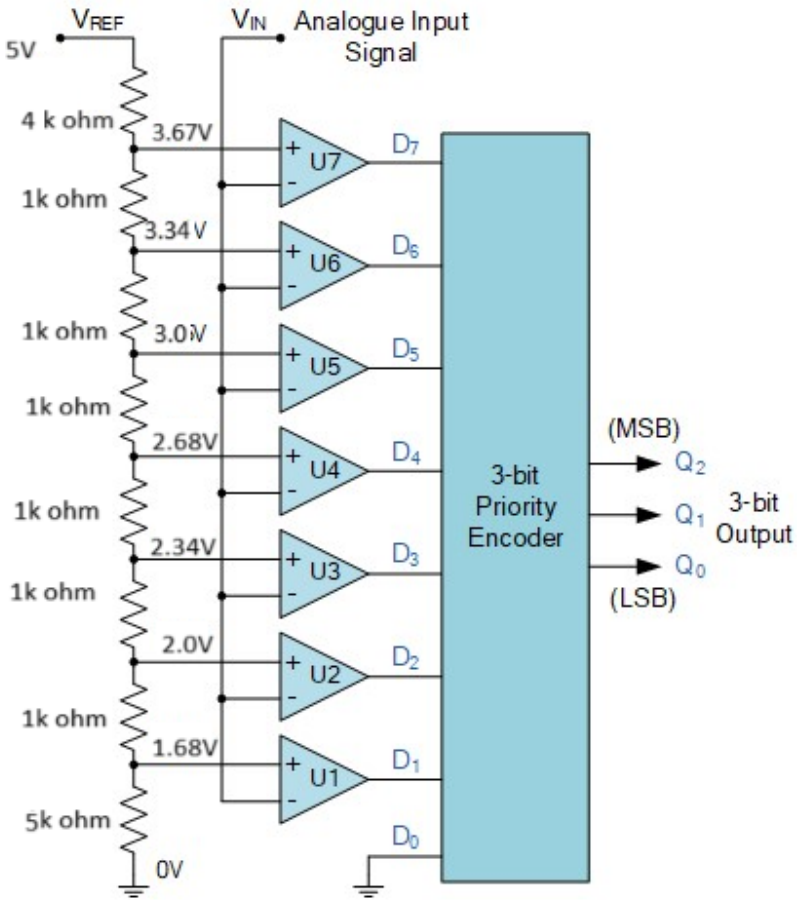
The priority encoder turns these comparator outputs into a binary number. It outputs the binary code corresponding to the highest comparator number that has a high output, effectively encoding the highest voltage level detected above the input voltage. voltage. The output D of the encoder is the binary representation of the highest active comparator:

$$D = Binary(i) \text{ where } C_i = 1 \text{ and } V_i + 1 = 0 \text{ (highest i with } C_i = 1$$

3 Truth Table

Analog Voltage	Comparator Outputs							Digital Outputs		
	D1	D2	D3	D4	D5	D6	D7	Q2	Q1	Q0
0 to 1.5	H	H	H	H	H	H	H	H	H	H
1.5 to 2	L	H	H	H	H	H	H	H	H	L
2 to 2.5	×	L	H	H	H	H	H	H	L	H
2.5 to 3	×	×	L	H	H	H	H	H	L	L
3 to 3.5	×	×	×	L	H	H	H	L	H	H
3.5 to 4	×	×	×	×	L	H	H	L	H	L
4 to 4.5	×	×	×	×	×	L	H	L	L	H
4.5 to 15	×	×	×	×	×	×	L	L	L	L

4 Schematic Diagram



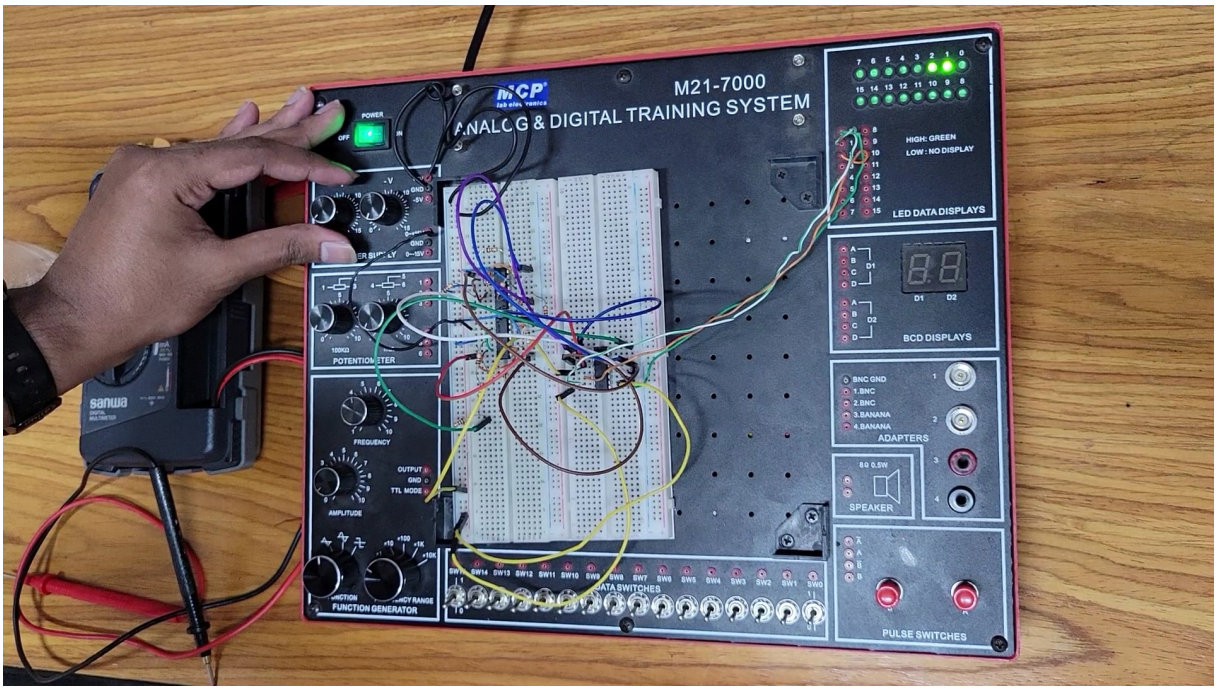
5 Apparatus

- **Operational Amplifiers (Op-Amps):** LM324N chips are acting as comparators. They check if the input voltage goes beyond specific levels.
- **Priority Encoder:** CD74HC147 is an 10-line to 4-bit active low priority encoder. It's like a translator, turning the signals from the comparators into a digital code. It picks the highest active signal and converts it into binary.
- **Resistors:**
  - $1k$ *Ohm* resistors are linked together to make a voltage divider. This setup helps create different reference voltages.
  - 470 *Ohm* resistors are linked to the LEDs
- **LED (Light Emitting Diode):** LEDs are used as indicators to show the digital output. They light up according to the binary code generated by the encoder.
- **Additional Materials:**
  - Jumper wires

- Breadboard
- Power Supply
- Variable Voltage Source

6 Implementation Steps

1. We connected eight $1k\ Ohm$ resistors in a row to make the resistor ladder on the breadboard.
2. Then, we linked the top of the ladder to the reference voltage source ($V_{ref} = 5V$) and connected the bottom to the ground.
3. We placed the LM324N in a way that each comparator's non-inverting input (+) was connected to a tap between resistors.
4. The inverting input (-) of all comparators was connected to a shared line where the input voltage (V_{in}, $0 - 7V$) would come in.
5. We positioned the CD74HC147 active low priority encoder close to the LM324N on the breadboard. We tied two of the highest priority inputs to High(as it is active low) and we are only considering the first three outputs to convert it into 8 to 3 bit encoder.
6. The outputs from the LM324N were then linked to the corresponding inputs on the CD74HC147.
7. We connected the 3-bit digital output from the CD74HC147 with LEDs.
8. The VCC pin of the LM324N was connected to a +5V power supply.
9. We made sure that the CD74HC147 also got power from $+5V$.
10. Before applying any input, we powered up the circuit and double-checked all connections using a multimeter.
11. Finally, we used a variable voltage source to give a variable voltage (ranging from 0 to $7V$ to V_{in} . We watched the digital output on an oscilloscope and confirmed that everything worked correctly across the input range.



7 Discussion

Developing and putting together our 3-bit Flash ADC project, we encountered some intricate technical hurdles. Particularly, we faced challenges with the trainer boards and noticed unexpected quirks with the breadboard. Tackling these obstacles became pivotal to ensure the smooth operation of our ADC circuit.

7.1 Troubleshooting

1. **Unexpected Breadboard Glitch:** Our breadboard had a glitch. Somehow, the positive and negative lines got mixed up, causing a short circuit. This dropped the voltage across our setup to just 0.6 volts, too low for our experiments. We used a multimeter to trace and fix the short circuit. Once resolved, our experiments continued with the correct voltage levels.

2. **Trainer Board Troubles:** We ran into trouble with our lab's trainer boards. Instead of behaving as expected, all the flash LEDs lit up simultaneously because of unusually high voltage outputs. Despite our efforts to troubleshoot and adjust settings, only the third flash LED didn't light up as needed for our project. After extensive testing, we discovered that only 3 trainer boards were working correctly. The others had the same issue, suggesting a broader problem with those specific boards.

3. **Voltage Measurement Accuracy:** We faced some issues with voltage measurements. If V_{ref} (reference voltage) was too high, it messed up the comparisons and made the results less accurate. Also, when V_{in} (input voltage) got too high, the output tended to stay stuck at the highest level most of the time.